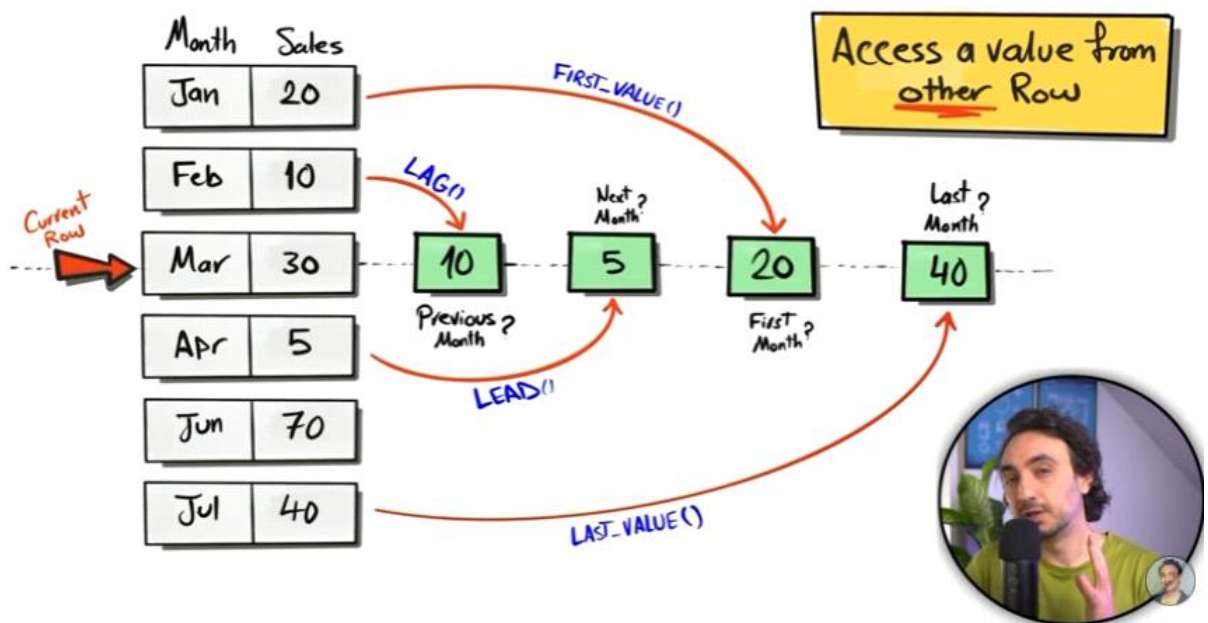


LEAD() LAG()
FIRST-VALUE() LAST-VALUE()

Value function :

Access value from another row.



ORDER BY clause is required

Function	Expression	Partition	Order	Frame
LEAD()	Any datatype	optional	required	Not allowed

Function	Expression	Partition	Order	Window
LEAD()	Any data type	Optional	required	Not allowed
LAG()	"	"	"	"
FIRST_VALUE()	"	"	"	Optional
LAST_VALUE()	"	"	"	Recommended

LEAD()

Access value from next row within a window

LAG()

Access value from previous row within a window

Syntax :

LEAD(Sales, 2, 10) OVER(ORDER BY <col-name>)

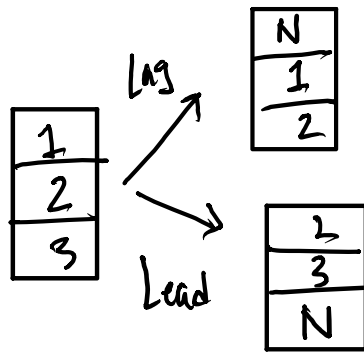
Expression
(required)

offset

default value
if no value is
found.

lag

N
1



Use case

Month-over-month performance.

Time-Series Analysis

Year-over-year
Overall performance.

Month-over-month.

Customer Retention Analysis

FIRST_VALUE()

Get first row in the window.

LAST_VALUE()

LAST-VALUE()

Get last row in the window.

Window definition is very important.

Default frame definition

ROWS BETWEEN UNBOUNDED PRECEDING AND
CURRENT ROW

works well for FIRST-VALUE().

Absolutely useless for LAST-VALUE()

For LAST-VALUE() we can use:

ROWS BETWEEN CURRENT ROW AND
UNBOUNDED FOLLOWING

To get the effect of LAST-VALUE(), FIRST-VALUE()
can be used while using descending order.

For numeric values, **MIN()**, **MAX()** functions can be used.

WINDOW VALUE (ANALYTICAL) FUNCTIONS

Allow Access specific Value from another Row



Rules

- Expression — **Any Data Type**
- ORDER BY — **Required**
- FRAME — **Optional**

Use Cases

- Time Series Analysis: **MoM + YoY**
- Time Gaps Analysis: Customer Retention
- Comparison Analysis: Extreme $\begin{cases} \text{Highest} \\ \text{lowest} \end{cases}$

